F-Distribution (Fisher–Snedecor)

F-distribution

- right-skewed distribution used most commonly in ANOVA (Analysis of Variance). - (https://en.wikipedia.org/wiki/F-distribution)
- named after Ronald Fisher and George W. Snedecor

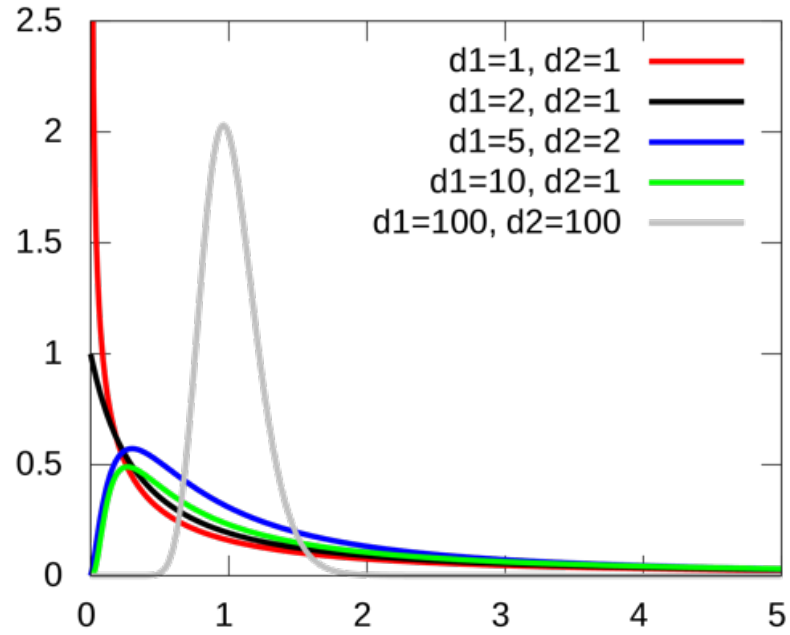
Ronald Fisher 1890-1962
British. Developed the
analysis of variance ANOVA

George W. Snedecor
1881-1974, USA

A random variate of the F-distribution

$$X = (U_1/d_1) / (U_2/d_2)$$

where U_1 and U_2 are independent and have chi-squared distributions with d_1 & d_2 degrees of freedom respectively.

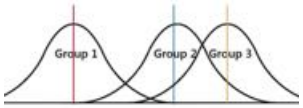


F-tests:

We have two "populations" of data.
They both have normal distributions.
Our Null hypothesis is that two independent normal variances are equal.

We observe sums of some appropriately selected squares are then examine them to see whether their ratio is significantly incompatible with this null hypothesis.

ANOVA = Analysis of Variance



Ronald Fisher 1890-1962 British. Developed the analysis of variance **ANOVA (1925)**

In its simplest form, **ANOVA** provides a statistical test of whether two or more population means are equal, and therefore generalizes the t-test beyond two means. The observed variance in a particular variable is partitioned into component.
- https://en.wikipedia.org/wiki/Analysis_of_variance

One-way ANOVA - compare means of two or more samples (using the F distribution). Response data "Y" (numerical) and input data "X" (numerical or categorical)

Two-way ANOVA is an extension of the one-way ANOVA that examines the influence of two different categorical independent variables (X1, X2) on one continuous dependent variable "Y". Goal - assess the main effect of each independent variable and assess if there is any interaction between them.

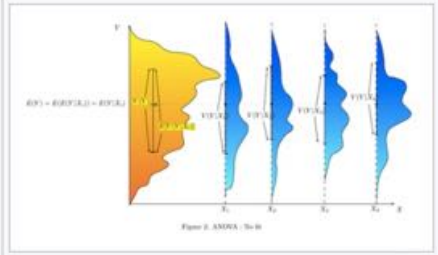
Variance Between

Variance Within

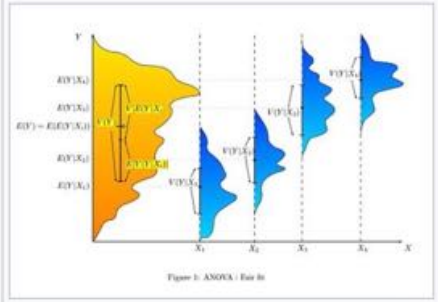
ANOVA is the synthesis of several ideas and it is used for multiple purposes.
As a consequence, it is difficult to define concisely or precisely.

"Classical" **ANOVA** for balanced data does three things at once:
1. Uses additive data decomposition for exploratory data analysis.
   Uses sums of squares to indicate the variance of each component of the decomposition.
2. Comparisons of mean squares, along with an F-test ... allow testing of a nested sequence of models.
3. Closely related to the **ANOVA** is a linear model fit with coefficient estimates and standard errors.

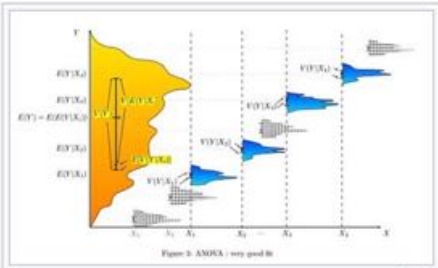
Additionally, **ANOVA** :
1. is computationally elegant and relatively robust against violations of its assumptions.
2. provides strong (multiple sample comparison) statistical analysis.
3. has been adapted to the analysis of a variety of experimental designs.



No fit: Young vs old, and short-haired vs long-haired



Fair fit: Pet vs Working breed and less athletic vs more athletic



Very good fit: Weight by breed

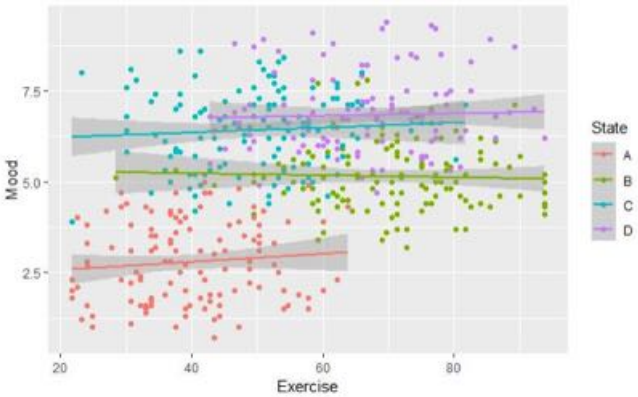
ANOVA = Analysis of Variance

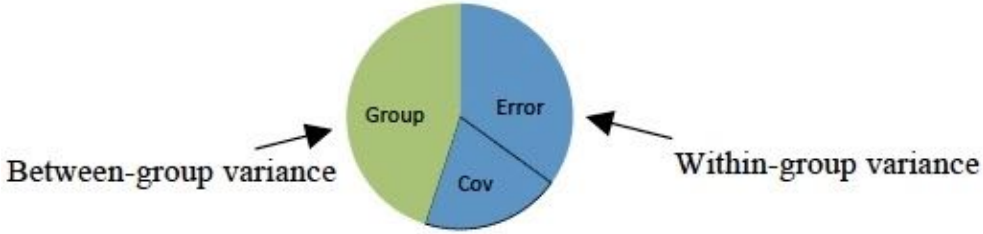
```
import numpy as np
from scipy import stats

import matplotlib.pyplot as plt
%config InlineBackend.figure_format = 'retina'

# Example: two-tailed test

alpha = 0.05

a = [1, 2,  1, 2, 1, 2,  1, 2,  1, 2]
b = [1, 3, -1, 2, 1, 5, -1, 6, -1, 2]

var_a = np.var(a, ddof=1)
var_b = np.var(b, ddof=1)
print(f"Variance a={var_a:.3f}, Variance b={var_b:.3f}")

f_stat = var_a/var_b # because we estimate mean from data

# build an F-distribution object
f_dist = stats.f(len(a)-1,len(b)-1)

f_crit_val = f_dist.ppf(0.95)
p_val = 2*min(f_dist.cdf(f_crit_val), 1-f_dist.cdf(f_crit_val))

print(f"f_crit_val = {f_crit_val:.4f}")
print(f"f_stat     = {f_stat:.4f} = var_a / var_b")
print(f"p_val      = {p_val:.4f}")

if (p_val < alpha):
    print(f"Reject H0")
else:
    print(f"Accept H0")
```

ANOVA – Multilevel model

https://en.wikipedia.org/wiki/Multilevel_model

Multilevel models, a.k.a.:
- hierarchical linear models
- linear mixed-effect model
- mixed models
- nested data models
- random coefficient
- random-effects models
- random parameter models
- split-plot designs





Multilevel models can be used as an alternative to **ANCOVA** (Analysis of Covariance).
ANCOVA blends ANOVA and regression.
ANCOVA evaluates whether the means of a dependent variable **"Y"**
are equal across levels of a categorical independent variable **"X"** often called a treatment,
while statistically controlling for the effects of other continuous variables (a.k.a. covariates **"CVs"** or nuisance variables).

Mathematically, **ANCOVA** decomposes the variance in the **"Y"** into variance explained by the **CVs**,
variance explained by the categorical **"X"**, and residual variance.

Intuitively, **ANCOVA** can be thought of as 'adjusting' the **"Y"** by the group means of the **CVs**.

Fisher Exact Test

From Fisher's book "The Design of Experiments" (1935)
Lady Tasting Tea example in 2nd chapter
Lady claimed that she ca distinguish between two types of cups of tea:

1. tea added first, then milk
2. milk added first, then tea

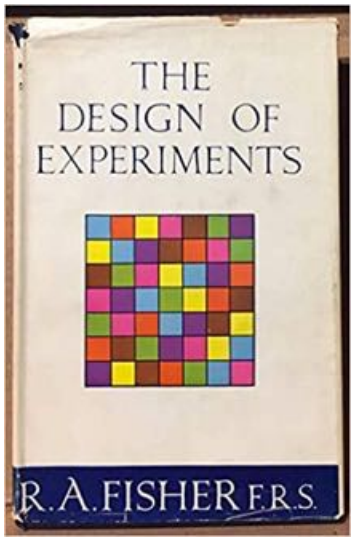
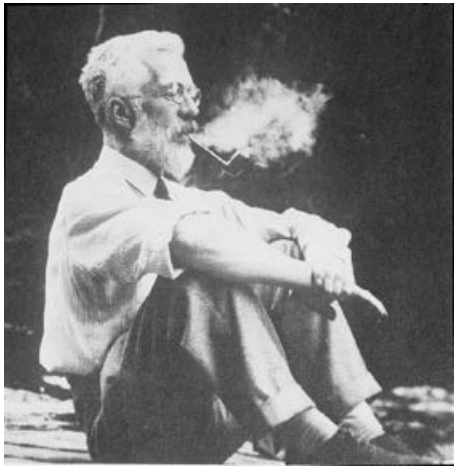
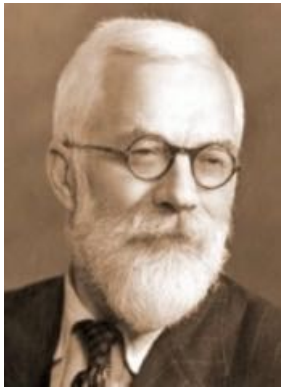
Author introduces the concept of "null hypothesis"

Experiment: given 4 cups of type 1 and 4 cups of type 2.
They look the same.
Lady tests them and labels the "milk first" ones.

The test is called "exact" because Fisher demonstrates how in this case he can calculate exact probability of each outcome.

If she guessed 3 out of 4, the probability is 16/70 = 0.22 – not good
If she guessed 4 out of 4 – the probability is 1/70 < 0.05 - good

Success count	Combinations	Number of Combinations
0	oooo	1 × 1 = 1
1	ooox, ooxo, oxoo, xooo	4 × 4 = 16
2	ooxx, oxox, oxxo, xoxo, xxoo, xoox	6 × 6 = 36
3	oxxx, xoxx, xxox, xxxo	4 × 4 = 16
4	xxxx	1 × 1 = 1
Total		70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